

University of California, Los Angeles

Predictive Modeling of Skin Cancer Diagnosis

Zoe Chan, Kaitlyn Wu, Jessica Xiao, Victoria Yee, Bebel Yen

STATS 101C

Professor Maierhofer

December 9, 2025

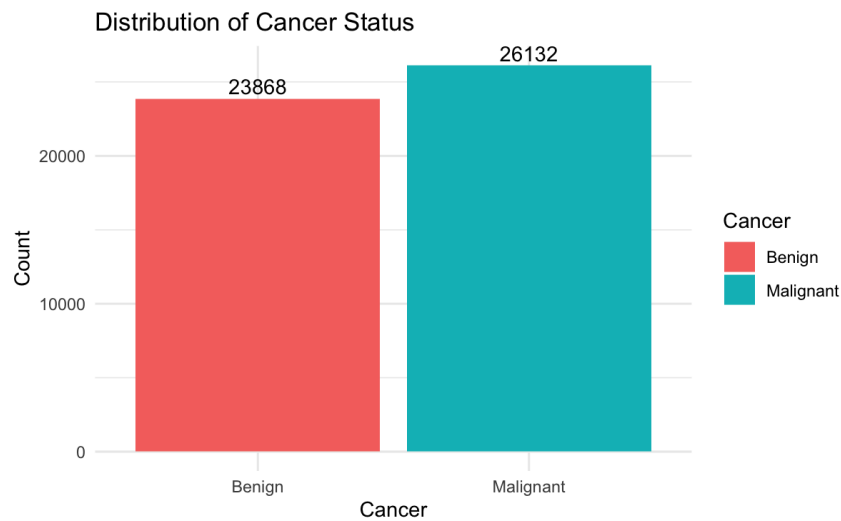
INTRODUCTION

In this project, we aim to predict the “cancer” status and the relation of 50 predictors to cancer risk of an individual based on one’s demographic, environmental factors, sun exposure, and other lifestyle behaviors.

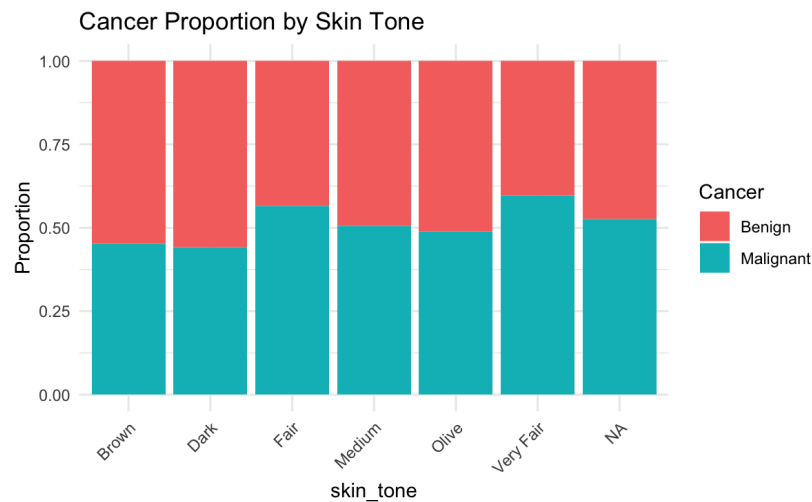
Among these variables, we believe that ultraviolet (UV) exposure is the primary cause of melanoma and other skin cancers, with indoor tanning before age 35 increasing melanoma risk by 59% (American Academy of Dermatology, 2024). Geographic location also contributes, as individuals in southern or high-altitude regions experience stronger UV radiation. Based on the background information provided, men face nearly double the melanoma mortality rate of women and racial disparities, with later diagnoses and poorer outcomes among people with darker skin (CDC, 2024). These background insights motivate our investigation of predictors of cancer status.

EXPLORATORY DATA ANALYSIS

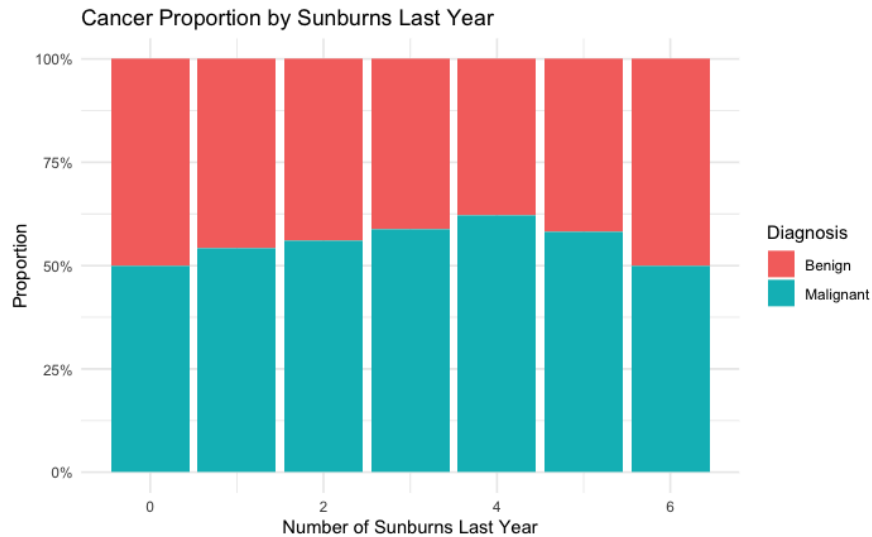
Below are plots visualizing relationships between different predictor variables.



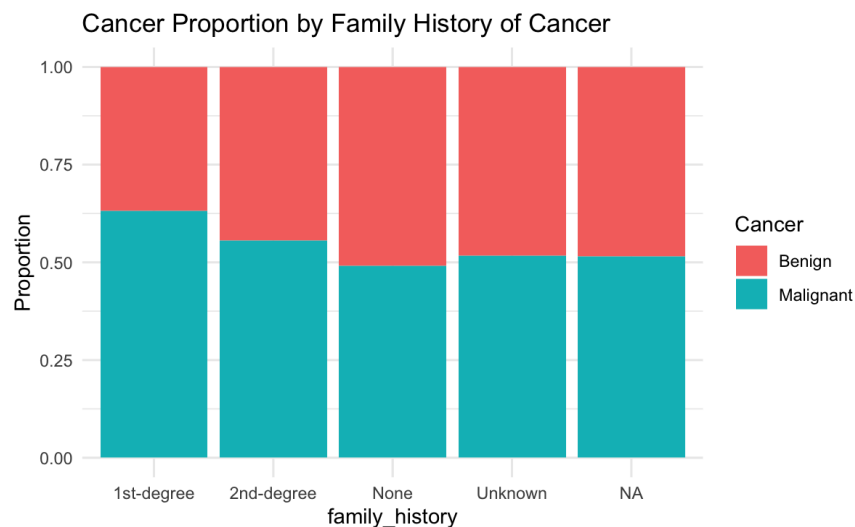
In total, there are 26,132 observations with a malignant cancer diagnosis and 23,868 observations with a benign cancer diagnosis in the dataset.



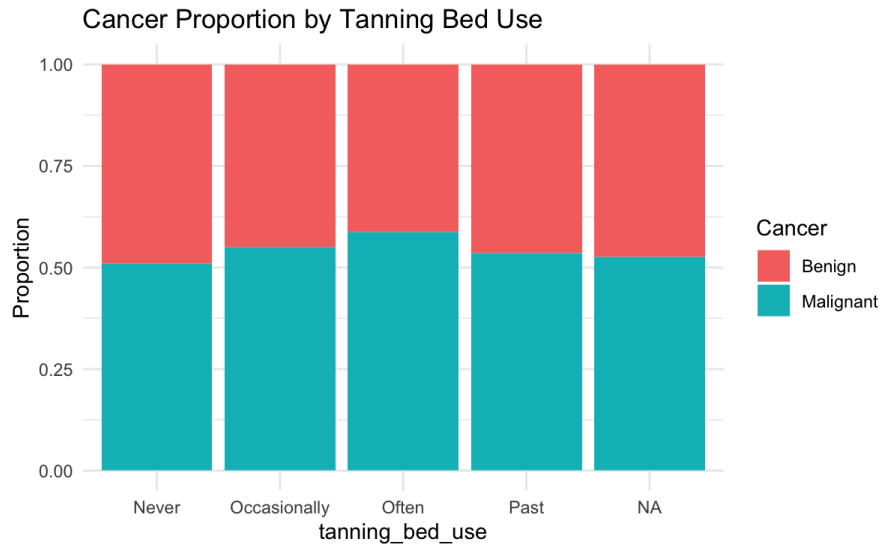
Those with Fair and Very Fair skin tone have a higher proportion of a malignant cancer diagnosis (around 55%-60%) compared to those with Brown or Dark skin tone (around 45%). Those with Brown or Dark skin tone have higher proportions of being diagnosed with benign cancer in comparison to those with Fair and Very Fair skin tones. In summary, individuals with tanner skin tones are less likely to have a malignant cancer diagnosis compared to those with lighter skin tones.



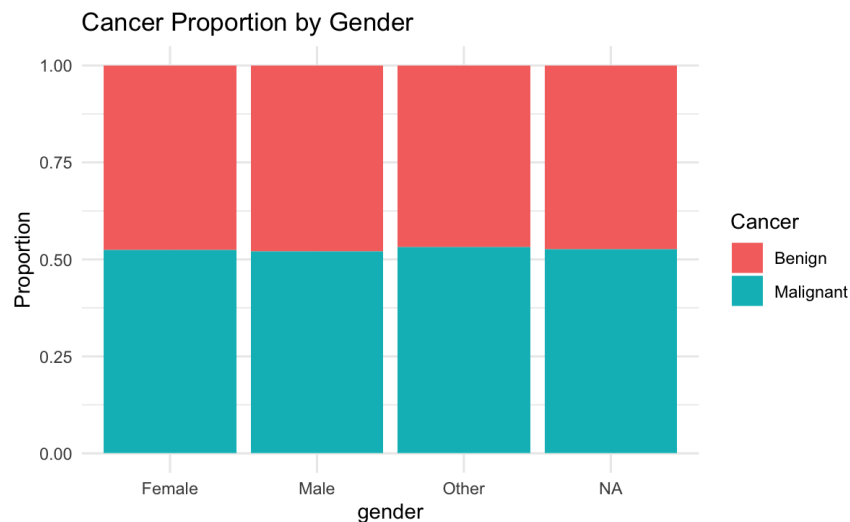
The visualization above shows that those with 3-5 sunburns in the previous year have the highest proportion of malignant cancer diagnoses and lowest proportion of benign cancer diagnoses, while those with 0 and 6 sunburns in the previous year have the highest proportion of benign cancer diagnoses and lowest proportion of malignant cancer diagnoses. While the pattern isn't perfectly linear, there is a general tendency for individuals with more sunburns to have a higher proportion of malignant diagnoses, especially between 1 and 4 sunburns.



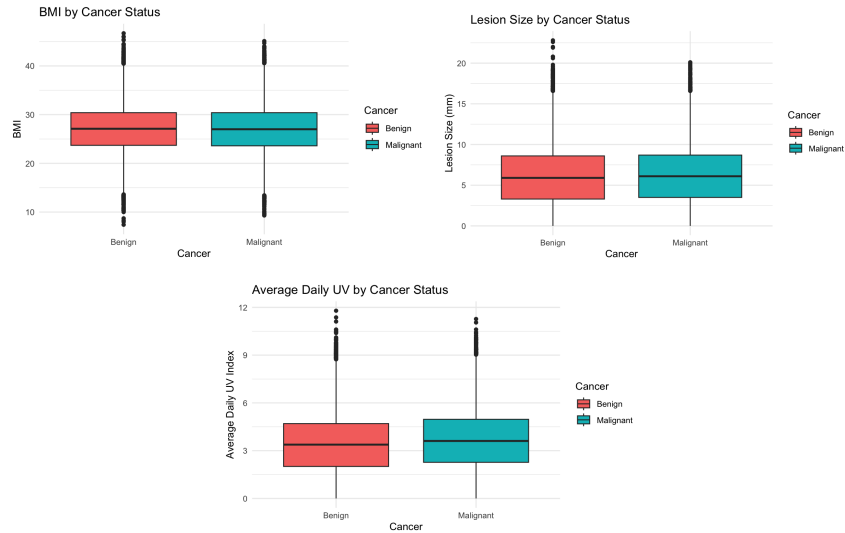
Individuals with a 1st-degree family history of cancer have the highest proportion of malignant skin cancer, with a percentage of around 63%. Those with no family history of cancer have the lowest proportion of malignant skin cancer, with a percentage of around 50%. Overall, the pattern suggests that having a closer family history of cancer (especially 1st-degree relatives) is associated with a higher likelihood of a malignant diagnosis. Individuals without a known family history tend to have more balanced benign and malignant proportions.



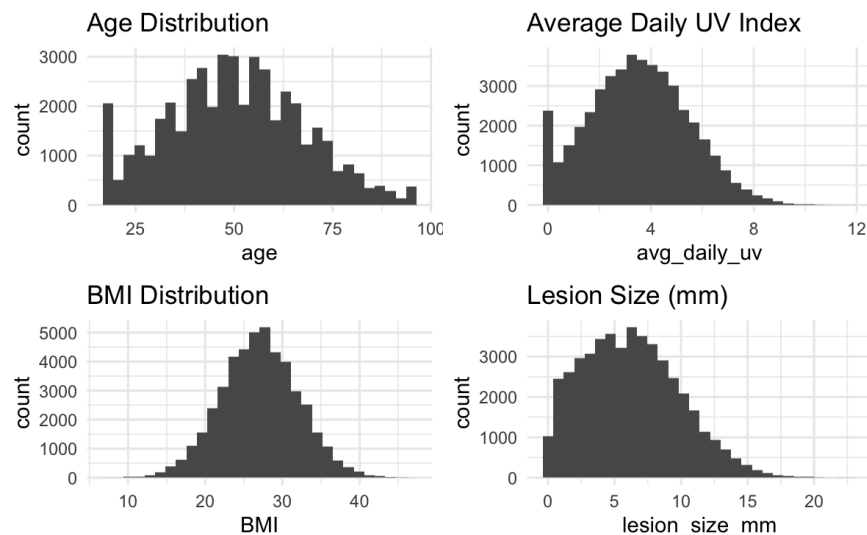
Individuals who often use tanning beds have the highest proportion of malignant cancer, with a percentage around 56%. Individuals who never use tanning beds have a similar proportion of malignant cancer to those who used tanning beds in the past, with a percentage around 50%-52%. While differences are not large, the pattern suggests that more frequent tanning bed use is associated with a higher proportion of malignant cancer diagnoses, particularly among individuals who report using tanning beds “often.” This aligns with existing understanding that artificial UV exposure can increase skin cancer risk.



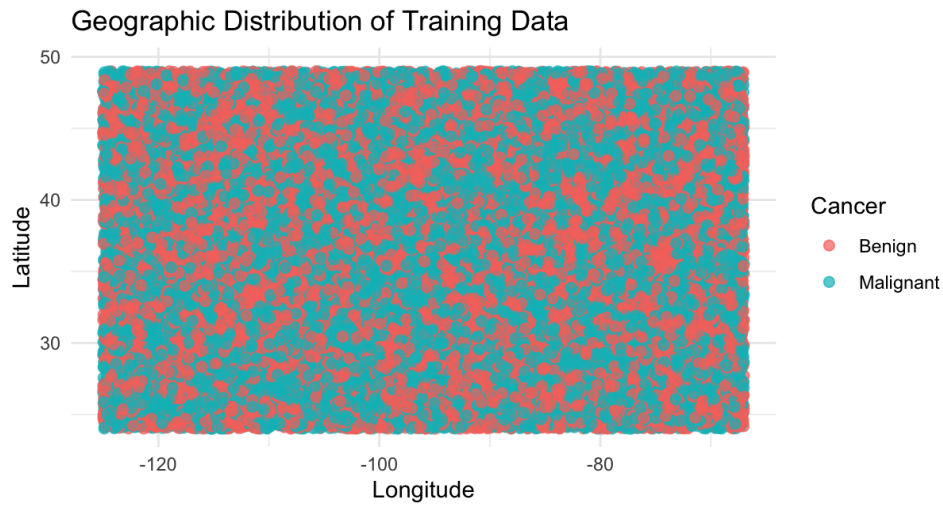
There is no significant difference in cancer proportion between females and males. Both have around a proportion of around 0.52 for malignant cancer and 0.48 for benign cancer. There is no clear relationship between gender and the likelihood of having a malignant vs. benign cancer diagnosis in this dataset. Gender does not appear to be a strong predictor of cancer outcome.



Across the three continuous predictors, lesion size and UV exposure show clearer associations with cancer status than BMI. BMI distributions for benign and malignant cases are nearly identical, suggesting that BMI is not a strong predictor. Malignant cases tend to exhibit slightly larger lesion sizes, with higher medians and more extreme upper-end outliers. Additionally, individuals with malignant diagnoses display slightly higher average daily UV exposure, with a higher median UV index compared to benign cases.



The distribution of ages range from low twenties to high nineties, with a majority of the observations being middle-aged individuals. Average daily UV index is roughly normally distributed and centered around level 4 UV index. BMI is also normally distributed and centered around a BMI of 27-28. Lesion size is right skewed, with most lesions ranging from around 2mm-7mm in size. To create these distributions, we used the [gridExtra](#) package to arrange the four plots in a 2x2 grid.



Although latitude and longitude are included in the dataset, exploratory analysis shows that the coordinates are uniformly distributed across the full bounding box of the continental United States (24° – 49° N, -125° – -67° W). This uniform pattern does not resemble real population distributions and suggests that the coordinates are synthetic rather than actual participant locations. Because they do not encode meaningful geographic risk information, latitude and longitude variables were dropped and excluded from modeling. We used the [maps](#) package to visualize the geographic distribution by plotting their latitude and longitude coordinates over a U.S. map.

PREPROCESSING

To prepare the dataset for analysis, we first removed a set of variables that were clearly unrelated to skin cancer risk and would only introduce noise into the analysis. Specifically, we excluded the following fields: ID, zip_code_last_digit, favorite_cuisine, music_genre, preferred_shoe_type, phone_brand, monthly_screen_time_minutes, uses_smartwatch, desk_height_cm, commute_minutes, participates_outdoor_sports, years_lived_at_address, pets, and favorite_color. These features reflect personal preferences, lifestyle habits, or administrative identifiers that have no biological or epidemiological connection to melanoma or other skin cancers.

After removing these irrelevant predictors, we converted character variables into categorical factors and evaluated missingness patterns. Next, we examined the latitude and longitude fields and determined that the coordinates were uniformly distributed across the entire U.S. bounding box, indicating synthetic rather than meaningful geographic information; therefore, these variables were excluded from modeling.

To address missing values, we applied median/mode imputation using statistics calculated only from the training data to prevent data leakage. Numeric predictors were imputed with the training-set medians, and categorical predictors were imputed with their most frequent category. We then aligned factor levels between the training and test sets and verified that no missing values remained.

The resulting cleaned dataset contains only relevant predictors, properly formatted and free of missing entries, providing us with a reliable foundation for exploratory analysis and classification modeling.

CANDIDATE MODELS

HYPERPARAMETER TUNING:

To improve predictive performance, we conducted systematic hyperparameter tuning for the Decision Tree, Random Forest, and XGBoost models. Tuning was performed using a grid search strategy. Each candidate model was fit on the training subset, and performance was evaluated using AUC (Area Under the ROC Curve) on the validation set. The hyperparameter set that achieved the highest validation AUC was selected as optimal and later used to refit the final model on the full training data.

Logistic Regression

Logistic regression does not contain tunable hyperparameters in its standard form, so no tuning was performed.

Decision Tree

We tuned two complexity parameters:

- `cp`: the complexity parameter controlling cost-complexity pruning
- `minsplit`: minimum number of samples required to attempt a split

Grid searched values:

- $cp \in \{0.001, 0.006, \dots, 0.05\}$
- $minsplit \in \{10, 20, 30, 40\}$

Random Forest

We tuned two parameters known to strongly influence random forest performance:

- `mtry`: number of randomly selected predictors available at each split
- `ntree`: total number of trees grown

Grid searched values:

- $mtry \in \{2, \sqrt{p}, p/3\}$ where p is the number of predictors
- $ntree \in \{200, 500, 800\}$

XGBoost

We tuned four of its most influential parameters:

- eta: learning rate
- max_depth: tree depth
- subsample: row sampling rate
- colsample_bytree: feature sampling rate

Grid searched values:

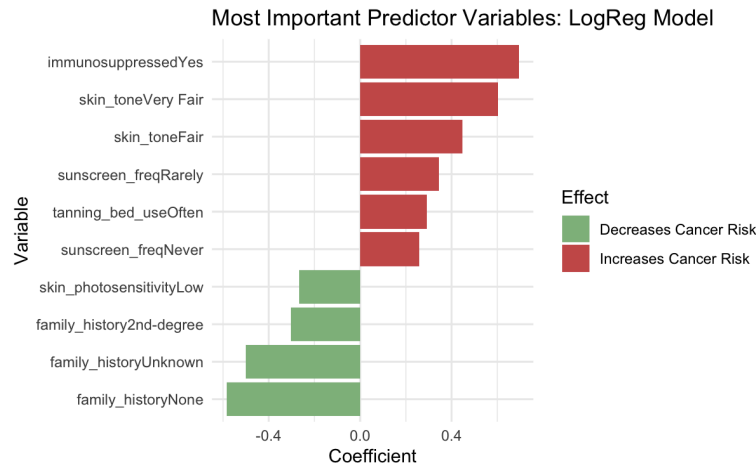
- $\text{eta} \in \{0.03, 0.05, 0.1\}$
- $\text{max_depth} \in \{3, 6, 9\}$
- $\text{subsample} \in \{0.6, 0.8, 1.0\}$
- $\text{colsample_bytree} \in \{0.6, 0.8, 1.0\}$

CANDIDATE MODELS:

Model ID	Model Type	Model Variables	Hyperparameters
M1	Logistic Regression	All variables after dropping unrelated predictors	No hyperparameters (standard GLM with binomial family)
M2	Decision Tree	All variables after dropping unrelated predictors	cp: 0.001 (complexity parameter), minsplit = 10
M3	Random Forest	All variables after dropping unrelated predictors	ntree = 800, mtry = 2
M4	XG Boost	All variables after dropping unrelated predictors	eta = 0.1 (learning rate), max_depth = 3, subsample = 1, colsample_bytree = 0.6

M1 Logistic Regression Model

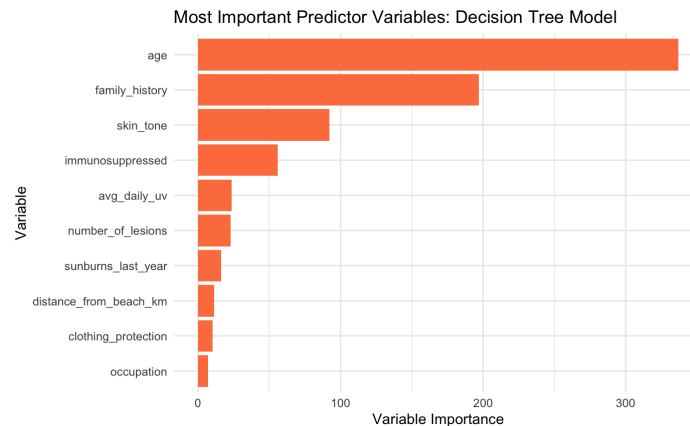
This model represents the log-odds of malignant cancer as a linear combination of our chosen predictor variables. It would include all (related) predictor variables and specifically uses a generalized linear model with a binomial family and logit link function. This was chosen as a candidate model because of its simplicity and speed.



For our logistic regression model, the most important predictor variables are immunosuppressive status, skin tone, and family history of cancer.

M2 Decision Tree Model

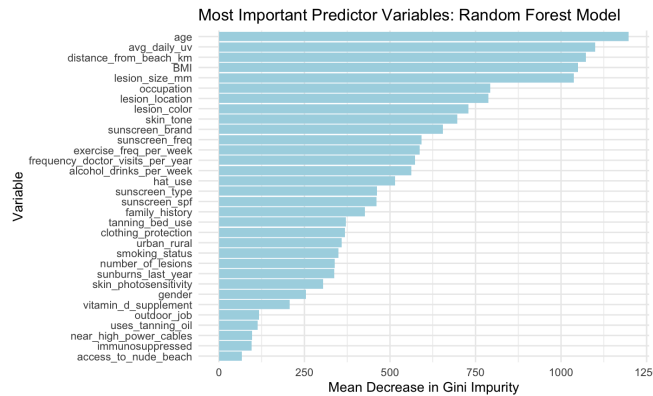
This model uses recursive binary splitting where at each node, a threshold is selected that separates benign diagnoses from malignant ones based on Gini impurity. This tree continues to split until it reaches our hyperparameter values (a complexity parameter of 0.01 and a minimum split size of 20). It was chosen as a candidate model because it identifies the most significant variables and can handle NA values, which this data has a lot of.



For our decision tree model, the most important predictor variables were age, family history of cancer, skin tone, and immunosuppressive status.

M3 Random Forest Model

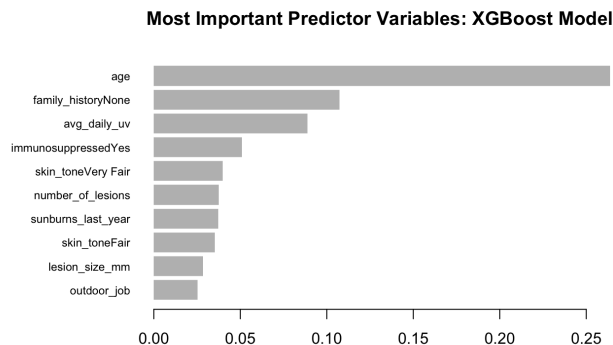
This model uses bagging and randomizes the variables used to create multiple decision trees. We decided on hyperparameter values of 500 trees and the square root of total features randomly sampled at each split. It was chosen as a candidate model because it's less prone to overfitting than singular decision trees and tends to have higher accuracy.



For our random forest model, the most important predictor variables are age, average daily UV, distance from a beach, BMI, and lesion size.

M4 XG Boost Model

This model utilizes boosting, or the sequential training of models based on previous models, on decision trees. We chose 100 trees as our number of boosting iterations, a learning rate parameter of 0.1, a max depth of 6, a proportion of randomly sampled training data per tree of 0.8, and a proportion of randomly sampled features per tree of 0.8. This was chosen as a candidate model because it should perform better than bagging and randomizing available features (random forest models).



Our XGBoost model shares the exact same important predictor variables as the random forest model.

Summary of Models

Given a patient who is immunosuppressed, fair-skinned, has a 1st-degree family history of cancer, 70+ years old with a high BMI, and gets high amounts of daily UV, all three models will flag them as high risk for cancer, but for different reasons. The logistic regression model will weigh immunosuppressive status, skin tone, and family history much more heavily than the random forest and XGBoost model. In contrast, those will weigh age, BMI, and average daily UV differently than the logistic regression model.

Example Cases

```
--- CASE 1 (ID: test_9290) ---
Logistic Regression: 42.5% Malignant → Benign
Random Forest:      44.4% Malignant → Benign
XGBoost:             41.0% Malignant → Benign
Ensemble (Average): 42.7% Malignant → Benign

--- CASE 2 (ID: test_18753) ---
Logistic Regression: 51.1% Malignant → Malignant
Random Forest:      51.0% Malignant → Malignant
XGBoost:             48.9% Malignant → Benign
Ensemble (Average): 49.9% Malignant → Benign
```

The model predictions for these two cases show the varying levels of diagnostic certainty. Case 1 (ID: test_9290) demonstrates strong model agreement, with all four models predicting “benign” approximately 42-44% malignancy probability, suggesting this patient has lower-risk characteristics such as smaller lesion size, reduced UV exposure, or absence of family history.

In contrast, Case 2 (ID: test_18753) represents a borderline case where the models produce conflicting predictions hovering around the 50% decision threshold. Two models classify it as malignant (51.1% and 51.0%) while two predict benign (48.9% and 49.9%). This near-equal split indicates the patient likely presents with mixed risk factors that make classification difficult.

MODEL EVALUATION:

Model Performance:

Model ID	Mean Accuracy	Mean Error	AUC
M1	0.6	0.2345	0.639
M2	0.5863	0.2413	0.6011
M3	0.5931	0.2378	0.6308
M4	0.5938	0.2363	0.6307

Model Performance (k-fold cross validation):

Model ID	Mean Accuracy	Mean Error	AUC
M1	0.5974	0.4026	0.6371
M2	0.5637	0.4363	0.5598
M3	0.5897	0.4103	0.6232
M4	0.5886	0.4114	0.6263

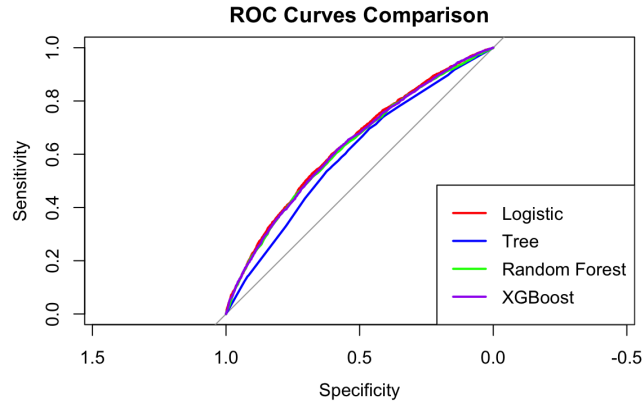
We used the [pROC](#) package for our validation set evaluation to compute ROC curves and AUC values. It allowed us to evaluate each model without having to convert each model to an mlr3 learner, which we would instead use later on in k-fold cross validation to handle the resampling loop.

Altogether, our four models have moderate predictive performance with AUC's ranging from 0.6011 to 0.639. The logistic regression model performs best out of the four, with the highest k-fold CV AUC of 0.6372 and a k-fold CV accuracy of approximately 59.83%.

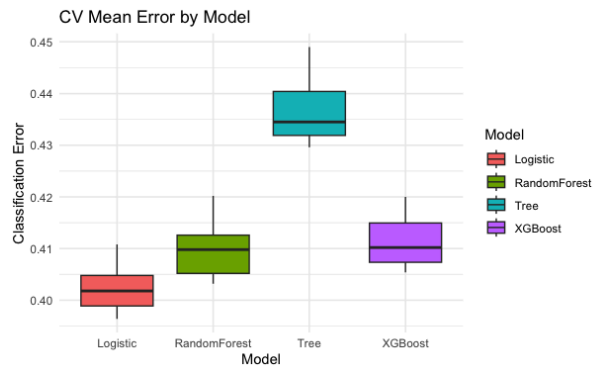
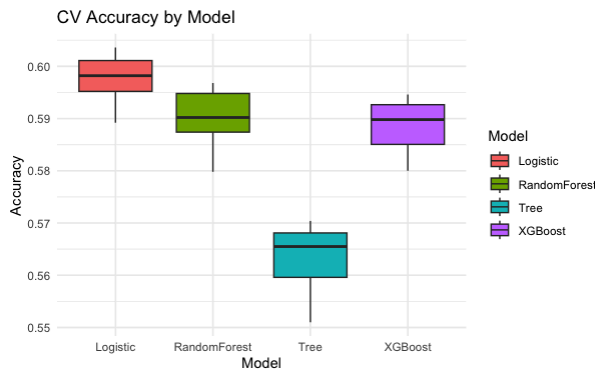
The random forest and XGBoost model have similar performance, with accuracies ranging from 56.37%-59.74% and AUC's ranging from 0.5598-0.6371.

The decision tree model performs worst of all four, with a k-fold CV AUC of 0.5637 and an accuracy of approximately 55.98%, which can be interpreted as slightly better than randomly guessing.

MODEL PERFORMANCE PLOTS:



The ROC comparison shows that Logistic Regression, Random Forest, and XGBoost all achieve very similar performance, with overlapping ROC curves and comparable AUC values. These 3 models clearly outperform the Decision Tree, whose ROC curve stays closer to the diagonal, indicating weaker predictive ability. This plot was created using the [pROC](#) package because of its built-in plotting functions that allowed us to easily visualize and compare model performance.



The CV Accuracy comparison shows that the Logistic Regression model achieves the highest accuracy with a mean of around 0.598. The Decision Tree model performs the worst and has the lowest accuracy with a mean of around 0.578. Random Forest and XGBoost models have similar accuracy scores of approximately 0.59.

The CV Mean Error comparison shows that the Logistic Regression model has the lowest average error rate of around 0.403. Again, the Random Forest and XGBoost model share similar mean error rates of approximately 0.41. The Decision Tree model's mean error of nearly 0.435 reinforces the idea that it is the worst performing model of the four.

FINAL DISCUSSION

Our final model was a logistic regression model fit on the full training data. We selected Logistic Regression as our final model because it achieved the best metrics—it had the highest accuracy, lowest mean error, and best AUC on our validation set among all candidate models. While Random Forest and XGBoost performed competitively, neither surpassed the logistic model on the key evaluation metrics used in this project.

Overall, the logistic regression model consistently achieved the highest accuracy scores and best AUC, finishing with an accuracy of 0.600 and an AUC of 0.639 and outperforming all of our other tree-based models. We validated this model through 10-fold cross-validation, which supported the idea that M1 held the best performance with an accuracy of 0.5974 and an AUC of 0.6371. It was able to generalize well to unseen data without overfitting, as shown through the consistency between regular validation performance and 10-fold cross validation performance.

Strengths of the logistic regression model are that it produces clear interpretations of probabilities, provides explicit odds ratios, and is overall a more computationally efficient method when compared to our other models. Its main limitations include linearity assumptions and limited flexibility in capturing nonlinear patterns.

Future improvements include feature engineering, regularized logistic regression, and expanded hyperparameter tuning. The collection of additional behavioral, environmental, or medical data would also be useful to improve predictive power.

APPENDIX: TEAM MEMBER CONTRIBUTIONS

- Zoe Chan: picked candidate models, evaluated models, and tuned models as needed
- Kaitlyn Wu: helped create and explain visualizations and initial data research
- Jessica Xiao: wrote discussion of final model, created candidate models and tables
- Victoria Yee: create visualizations and initial exploratory data analysis
- Bebel Yen: assisted with preprocessing data and choosing which models to use